

```

{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "10-07-2026_21:54:04",
  "variant" : 1,
  "subjectName" : "mathematics",
  "categories" : [ "Combinatorics", "Mathematical Logic", "Probability" ],
  "questions" : [ {
    "question" : "What is the value of the combinatorial coefficient  $C(5, 2)$ ?",
    "answers" : [ "5", "10", "20", "15" ],
    "correctIndex" : 1
  }, {
    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "32", "16", "25", "64" ],
    "correctIndex" : 0
  }, {
    "question" : "Which principle is also known as the Dirichlet Box Principle?",
    "answers" : [ "Inclusion-Exclusion Principle", "Multiplication Principle", "Pigeonhole Principle", "Bijection Principle" ],
    "correctIndex" : 2
  }, {
    "question" : "Which logical connective represents an 'exclusive or' (XOR)?",
    "answers" : [ "Both inputs are true", "Both inputs are false", "At least one input is true", "Exactly one input is true" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the negation of the statement 'All primes are odd'?",
    "answers" : [ "All primes are even", "Some primes are odd", "No primes are odd", "There exists at least one prime that is not odd" ],
    "correctIndex" : 3
  }, {
    "question" : "If P is true and Q is false, what is the truth value of the implication  $P \rightarrow Q$ ?",
    "answers" : [ "Contingent", "False", "Undefined", "True" ],
    "correctIndex" : 1
  }, {
    "question" : "If events A and B are independent, what is  $P(A \text{ and } B)$ ?",
    "answers" : [ " $P(A) + P(B)$ ", " $P(A) / P(B)$ ", " $P(A) * P(B)$ ", " $P(A) + P(B) - P(A \text{ or } B)$ " ],
    "correctIndex" : 2
  }, {
    "question" : "What does Bayes' Theorem calculate?",
    "answers" : [ "Marginal probability", "Conditional probability", "Joint probability", "Discrete distributions" ],
    "correctIndex" : 1
  }
]

```

```
}, {  
  "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a  
sample space?",  
  "answers" : [ "0", "100", "Infinite", "1" ],  
  "correctIndex" : 3  
}, {  
  "question" : "What is the probability of flipping a fair coin three times and getting all heads?",  
  "answers" : [ "1/6", "1/8", "1/4", "1/2" ],  
  "correctIndex" : 1  
}  
}]  
}
```